

Database Design Guide

Relational Databases

Relational databases store data in tables with rows and columns. SQL is the standard language for querying relational data. Primary keys uniquely identify rows. Foreign keys create relationships between tables. ACID properties ensure data integrity in transactions.

NoSQL Databases

NoSQL databases offer flexible schemas for unstructured data. MongoDB stores documents as JSON-like objects. Redis is an in-memory key-value store for caching. Cassandra handles large-scale distributed data. Choose NoSQL when you need horizontal scaling or flexible schemas.

Indexing Strategies

Indexes speed up query performance by creating lookup structures. B-tree indexes work well for range queries. Hash indexes are fast for equality lookups. Composite indexes cover multiple columns. Too many indexes slow down write operations. Analyze query patterns before adding indexes.

Query Optimization

Use EXPLAIN to analyze query execution plans. Avoid SELECT * and fetch only needed columns. Use parameterized queries to prevent SQL injection. Connection pooling reduces overhead of creating new connections. Batch inserts are faster than individual row inserts.

Database Security

Never store plain text passwords in databases. Use bcrypt or Argon2 for password hashing. Implement row-level security for multi-tenant applications. Encrypt sensitive data at rest and in transit. Regular backups protect against data loss. Audit logs track database access.